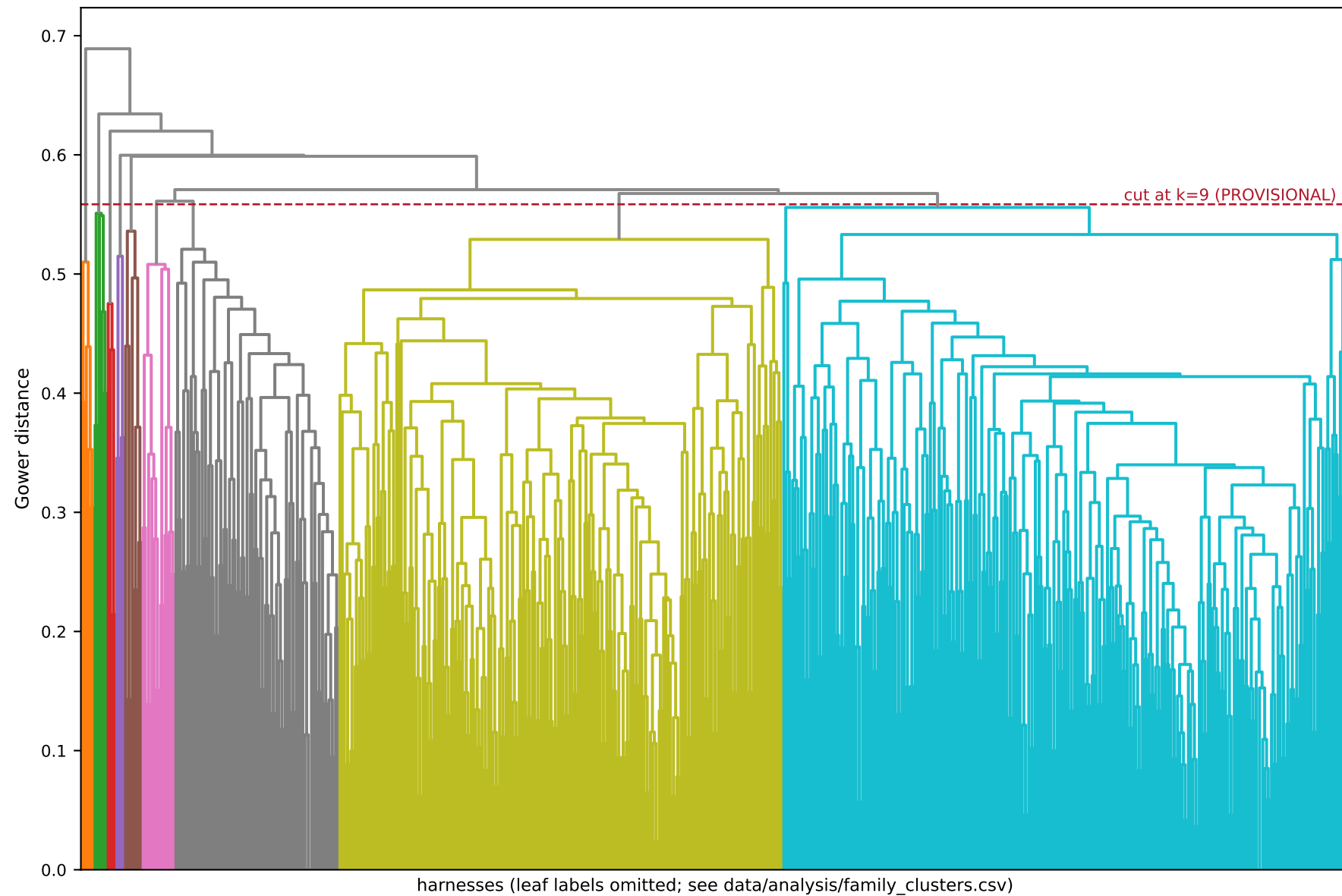
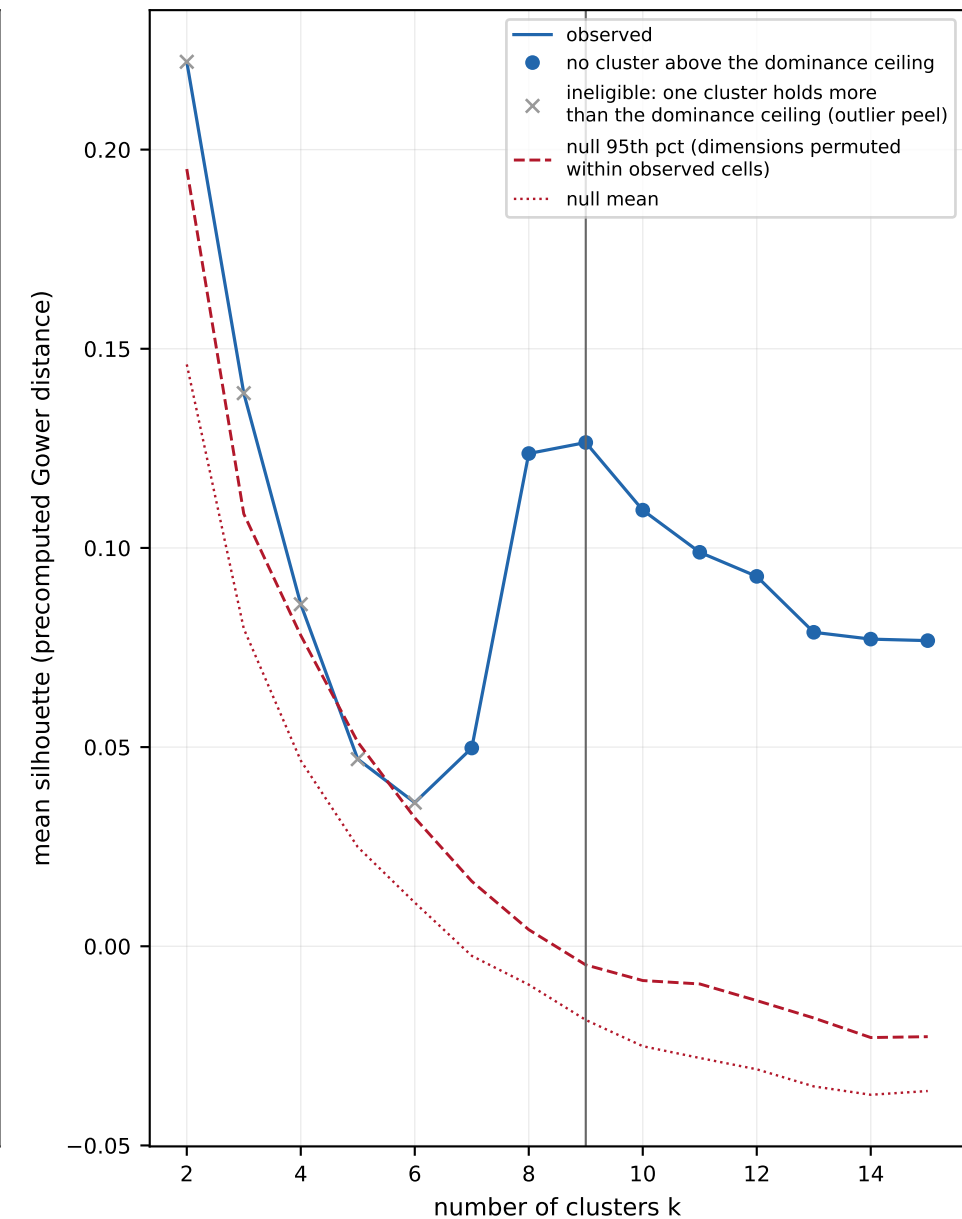


Average-linkage (UPGMA) dendrogram, Gower distance on coded cells only
580 systems, 37 dimensions, cophenetic $r = 0.57$



Families NOT supported (negative result)
best $k=9$, silhouette 0.126, stability 0.55



Gower distance averages, for each pair of systems, only the dimensions BOTH systems report: not_reported contributes nothing and unresolved cells are excluded, so shared silence cannot make two harnesses look alike. The null preserves every dimension's marginal distribution and every system's missingness pattern while destroying cross-dimension association. Rendered by scripts/analyse_families.py.